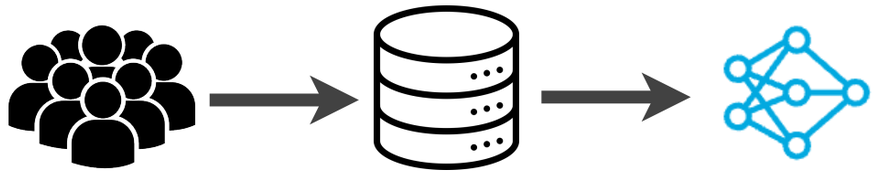
Frontiers of Offline Interactive Machine Learning: From Contextual Bandits to LLM Alignment

Kwang-Sung Jun (전광성)
Associate Professor

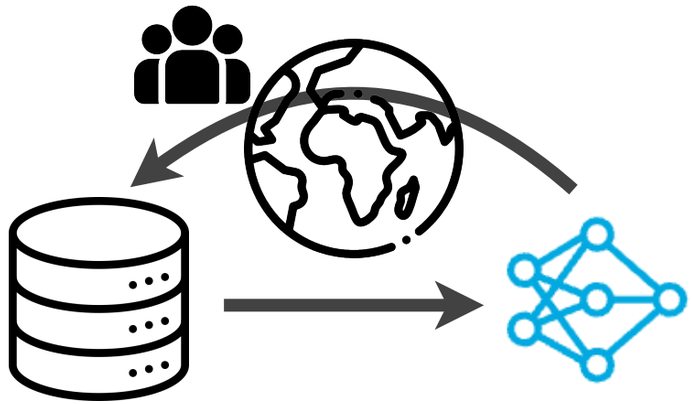
Department of Computer Science and Engineering
Graduate School of Artificial Intelligence
POSTECH

LeT Workshop
October 8, 2026

Supervised learning requires *curating* train data..



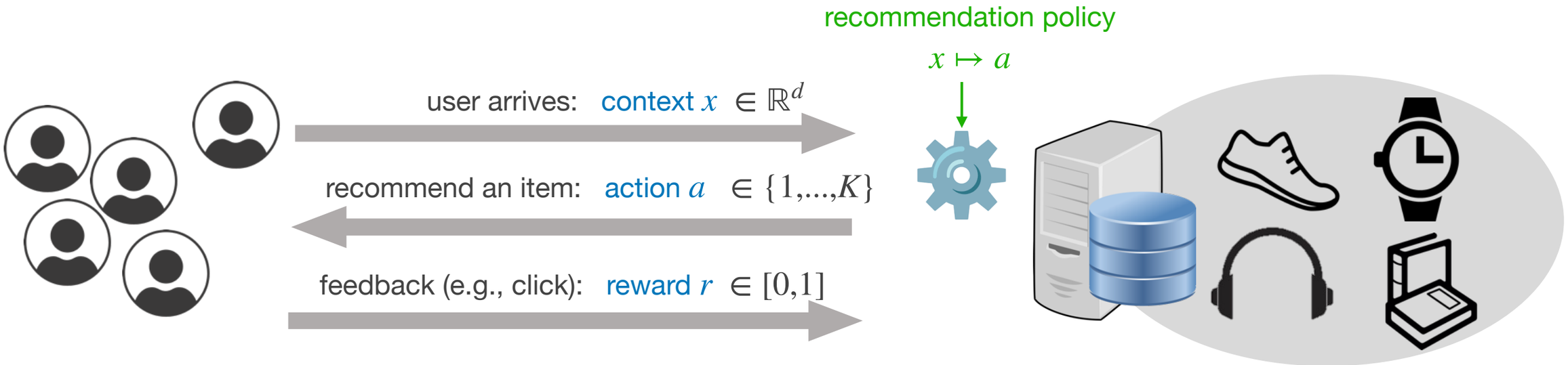
Interactive Machine Learning



agents that not only learn from data
but make data collection decisions!

Lower cost and time!

Example: Recommendation system

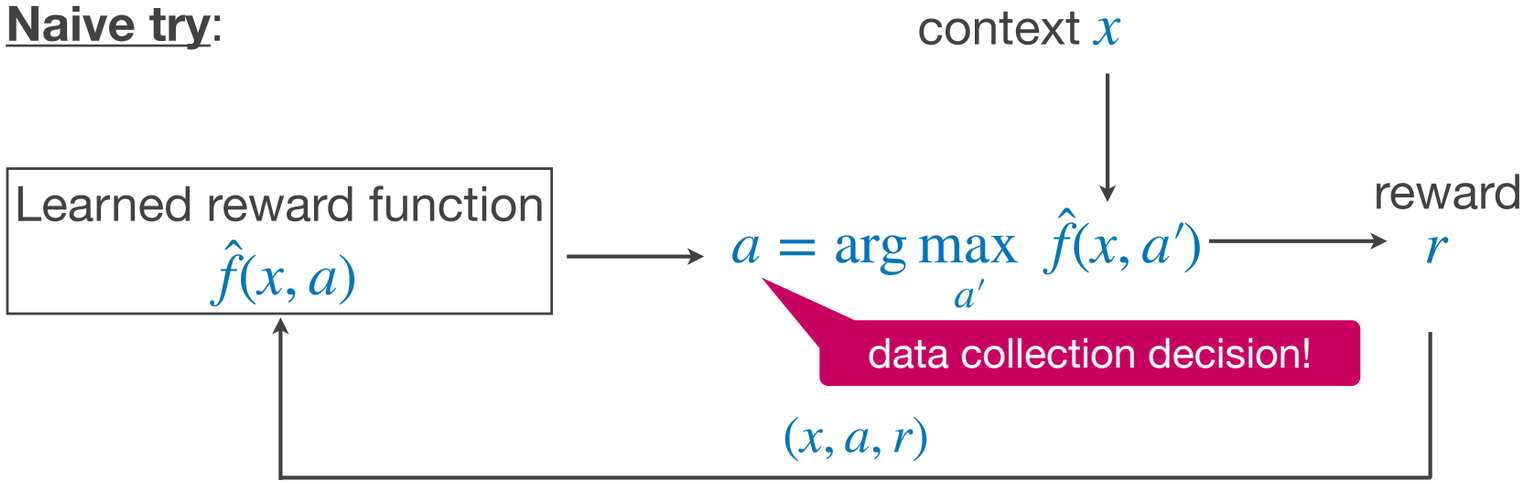


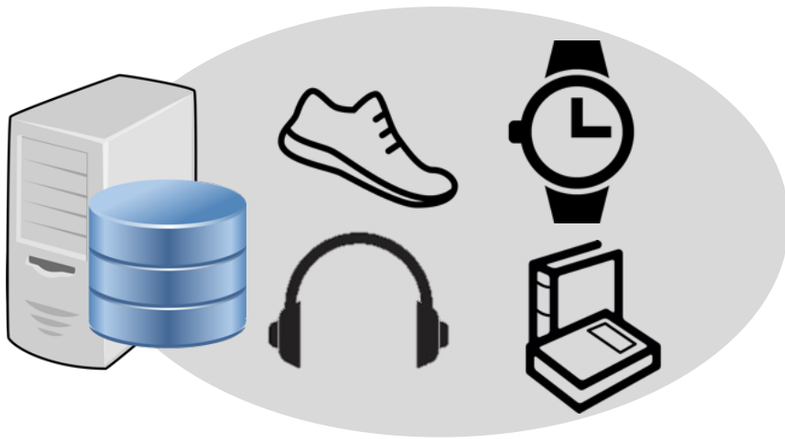
Goal: Maximize cumulative rewards

"contextual bandit" problem

The challenge of interactive ML

Naive try:



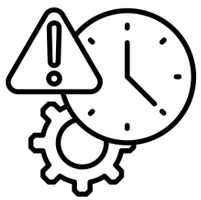


Challenge: The data (x, a, r) depends on $\hat{f}$. $\Rightarrow$ Feedback starvation on some actions.

Without proper exploration, the performance plateaus!

- A large body of literature on how to explore!
- **Implicit assumption** in most papers: online learning is possible

Online learning is usually not possible!



1. System constraints

- Recommendations must be real-time
- Good luck fighting with system engineers



2. Organizational policies

- Companies often have an approval process for new recommendation policies

  ⇒ Online learning is impossible!

Let’s go offline!



- Update the policy weekly.
- Use data collected by this week’s policy to train the next week’s policy ⇒ repeat!
- For simplicity, let’s focus on updating just once.

Offline contextual bandits

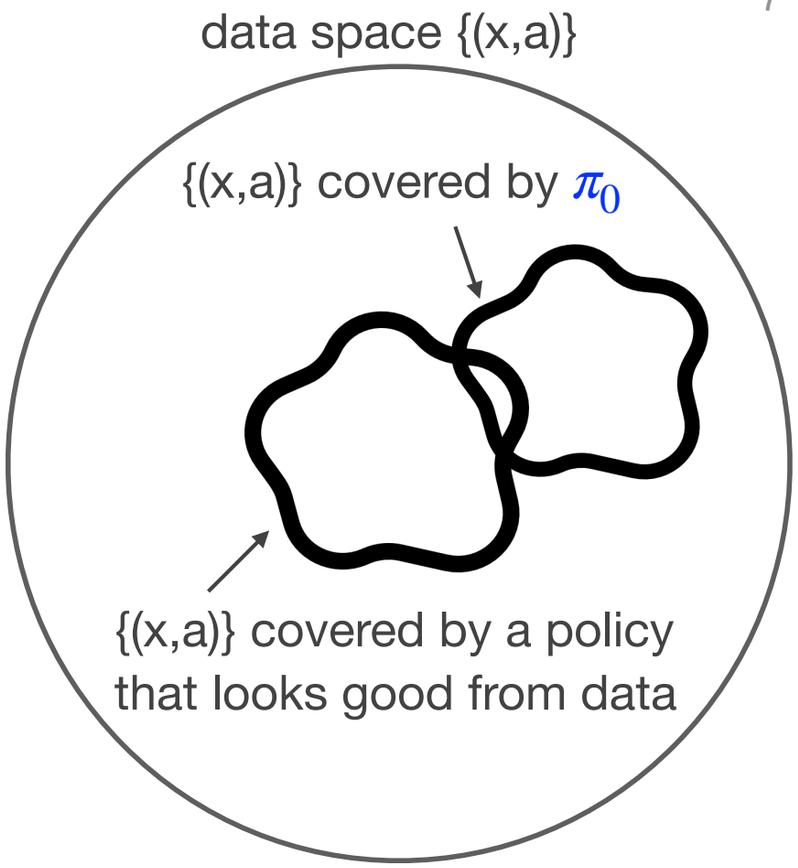
Policy execution:

For $t = 1, \ldots, n$

- Observe a **context** $x_t \sim \mathcal{D}$
- Choose an **action** $a_t \sim \pi_0(x_t)$ // behavior policy (distribution)
- Observe **reward** r_t

Given: $\{(x_t, a_t, r_t)\}_{t=1}^{n}$

Goal: Find the policy $\hat{\pi}$ that maximizes expected reward!



Key: should not deviate too much from π_0!
but what's the right measure?

The challenge: The data is **biased**!! ⇒ Risk of "over-optimization"

Part 1: Offline contextual bandits

The problem definition

Policy execution:

For $t = 1, \ldots, n$

- Observe a **context** $x_t \sim \mathcal{D}$
- Choose an **action** $a_t \sim \pi_0(x_t)$ // behavior policy
- Observe **reward** $r_t \in [0,1]$

Given: $D_n = \{(x_t, a_t, r_t)\}_{t=1}^{n}$

Π: Policy class (e.g., neural network architecture)

Goal: find $\pi^* = \arg\max_{\pi \in \Pi} \left[v(\pi) := \mathbb{E}_{x \sim \mathcal{D}, a \sim \pi(x)}[\text{reward}(x, a)] \right]$

↑ value of policy π

Parallel with supervised learning

	learning (=optimization)	**selection**
data split used	train set	validation set
# of models to choose from	∞	$< \infty$

---

classification task: minimize 0/1 ~~loss~~ ogate loss → ??

minimize 0/1 loss → ??

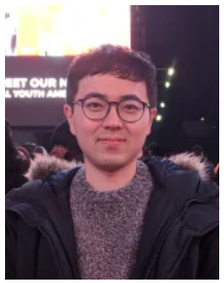
far from being solved!

Today's focus

[COLT'25, RyuKK**J**]

Improved Offline Contextual Bandits with Second-Order Bounds: Betting and Freezing

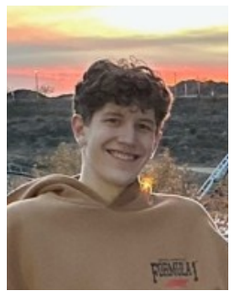
Joint work with:



Jongha Ryu
MIT (⇒ PSU)



Jeongyeol Kwon
UW-Madison (⇒ Meta)



Benjamin Koppe
Cornell

Selection: Why so hard?

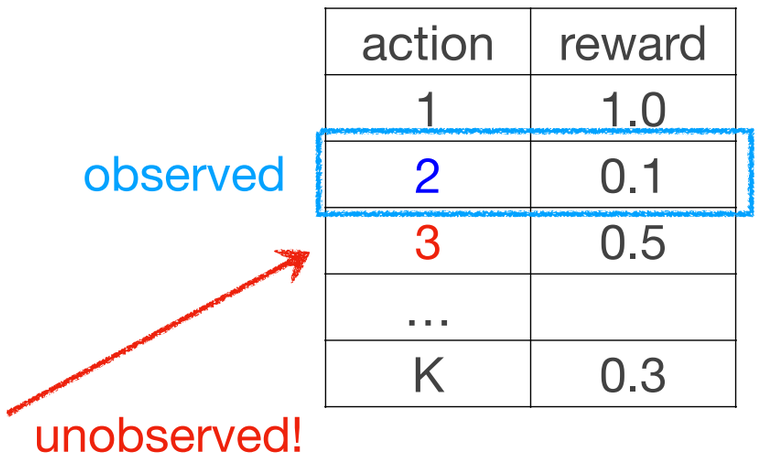
If we could estimate $v(\pi)$, then we could do argmax.

Task: estimate $v(\pi)$ using the data collected by the behavior policy π_0

At time t, the context was x_t

the behavior policy π_0 chose action 2

the target policy π chooses action 3



Standard solution: Use inverse propensity score (IPS)! $\mathrm{IPS}(\pi) = \frac{1}{n}\sum_{t=1}^{n} \tilde{r}_t(\pi)$ where $\tilde{r}_t(\pi) = \frac{\pi(a_t \mid x_t)}{\pi_0(a_t \mid x_t)} r_t$

unbiased!

Observation: high variance when π is very different from π_0

requirement: the behavior policy is stochastic

Challenges of IPS

Π: The set of policies to select from

Naive: $\max_{\pi \in \Pi} \mathrm{IPS}(\pi)$

may choose a highly-overrepresented policy

($\because$ its IPS has large variance)

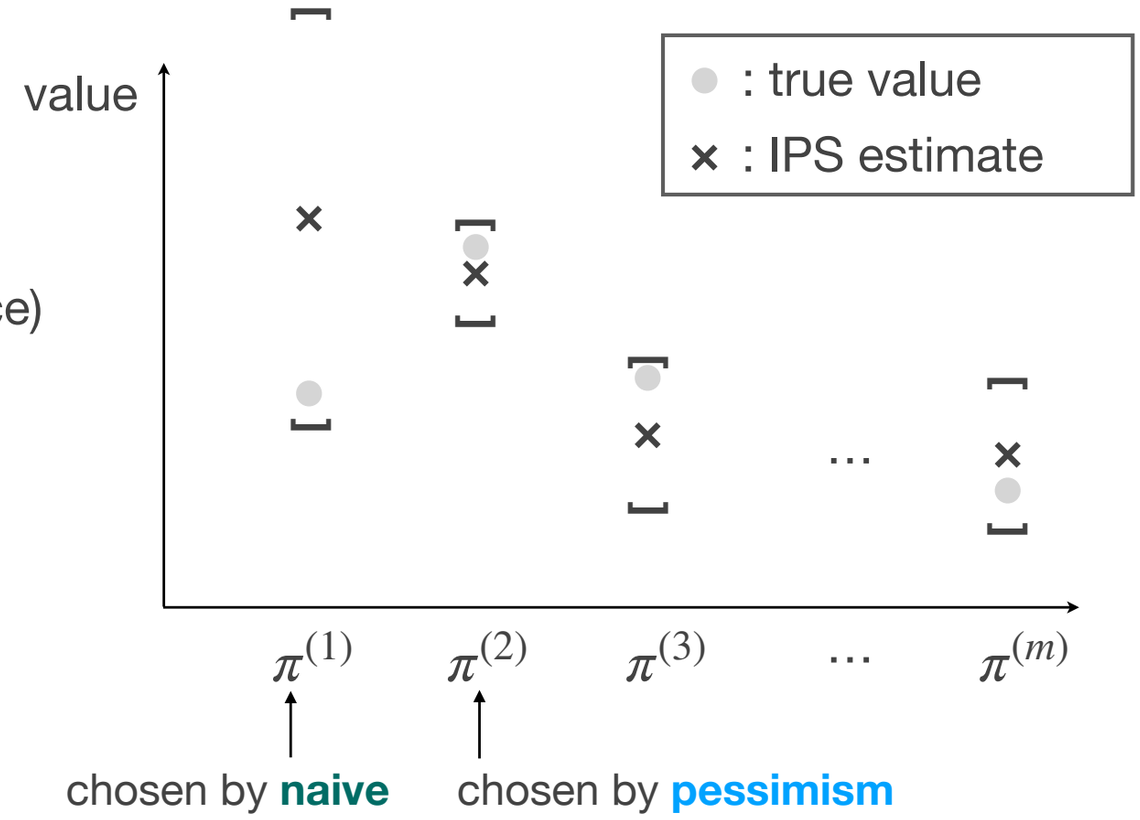
unstable!

Pessimism: use lower confidence bound (LCB) of $v(\pi)$

$\max_{\pi \in \Pi} \mathrm{LCB}(\pi)$

not swayed by highly-overrepresented policies

stable!



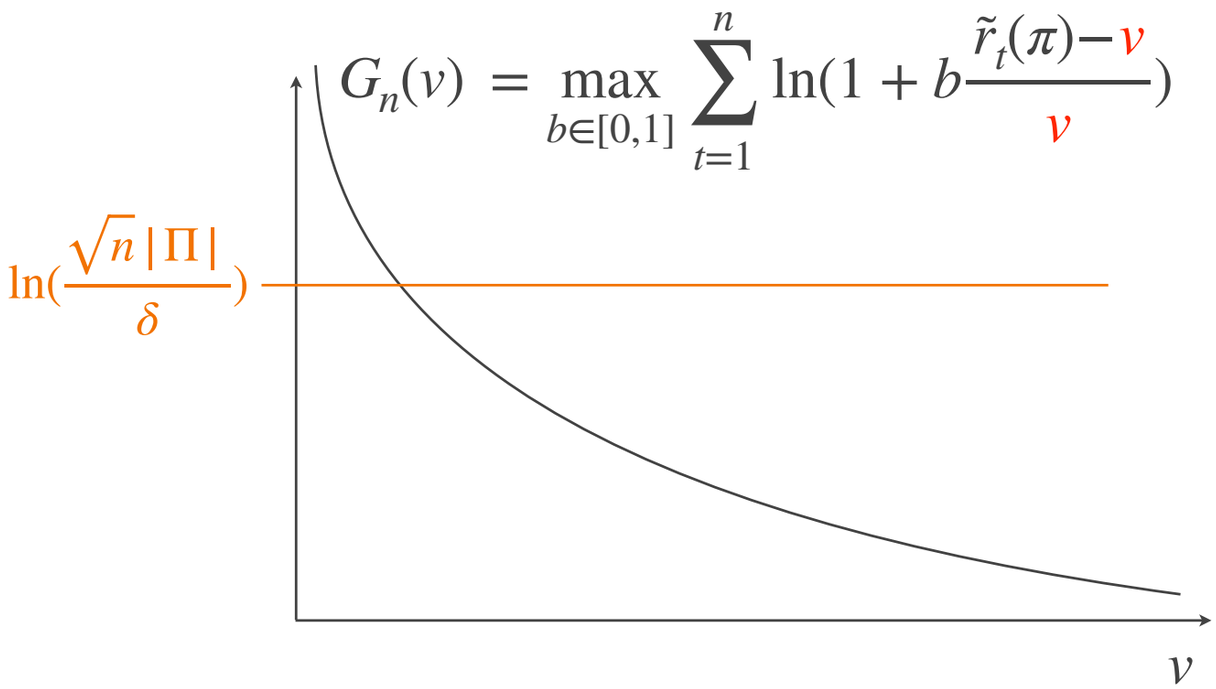
Our contribution:

- Provably tighter LCB than prior art ⇒ provably better selection!
- **Variance-adaptive**: Without knowing the variance, performs as well as knowing the variance!

PUB (Pessimism by semi-Unbounded coin Betting)

1. Find $\hat{v}$ such that $G_n(\hat{v}) = \ln(\frac{\sqrt{n}\,|\Pi|}{\delta})$

2. Report $\hat{v}$ as the LCB of $v(\pi)$.



Technical remark: $G_n(v)$ is the maximum log wealth achievable from a certain gambling

PUB Guarantee

Proposed: $\hat{\pi} = \arg\max_{\pi \in \Pi} \mathrm{LCB}_n^{\mathrm{PUB}}(\pi)$

PUB leads to an improved theoretical guarantee:

a measure of dist(π^*, π_0)

$$\mathbb{E}[\text{reward of } \pi^*] - \mathbb{E}[\text{reward of } \hat{\pi}] \lesssim \sqrt{\mathrm{Var}(\tilde{r}(\pi^*))\frac{\ln(|\Pi|/\delta)}{n}}$$

$$\tilde{r}(\pi^*) := \frac{\pi^*(a\,|\,x)}{\pi_0(a\,|\,x)} r(x,a) \in [0,\infty)$$

Naive: $\sqrt{\max_{\pi \in \Pi} \mathrm{Var}(\tilde{r}(\pi))\frac{\ln(|\Pi|/\delta)}{n}}$

Prior art [Sakhi+24]: $(1 + \mathbb{E}[\tilde{r}(\pi^*)^2])\sqrt{\frac{\ln(|\Pi|/\delta)}{n}}$

$$\sqrt{1 + \mathrm{Var}(\tilde{r}(\pi^*))} \;\le\; 1 + \mathrm{Var}(\tilde{r}(\pi^*)) \;\le\; 1 + \mathbb{E}[\tilde{r}(\pi^*)^2]$$

⇒ Two-step improvement!

Our contribution is more fundamental

- Our LCB is nonasymptotic and works for $[0,\infty)$-valued random variables
  - most prior work: asymptotic, or $[0,U]$-valued random variables
  - is a significant improvement from existing ones, both in theory and practice

- Our LCB is based on a connection between **online learning** and **concentration inequalities**.



Existence of an online learning algorithm with a good performance guarantee

implies

[× ●]

A tight, computable confidence bound

*Track record of tight confidence bounds using online learning (also invited to ICCOPT, JSM, etc.)

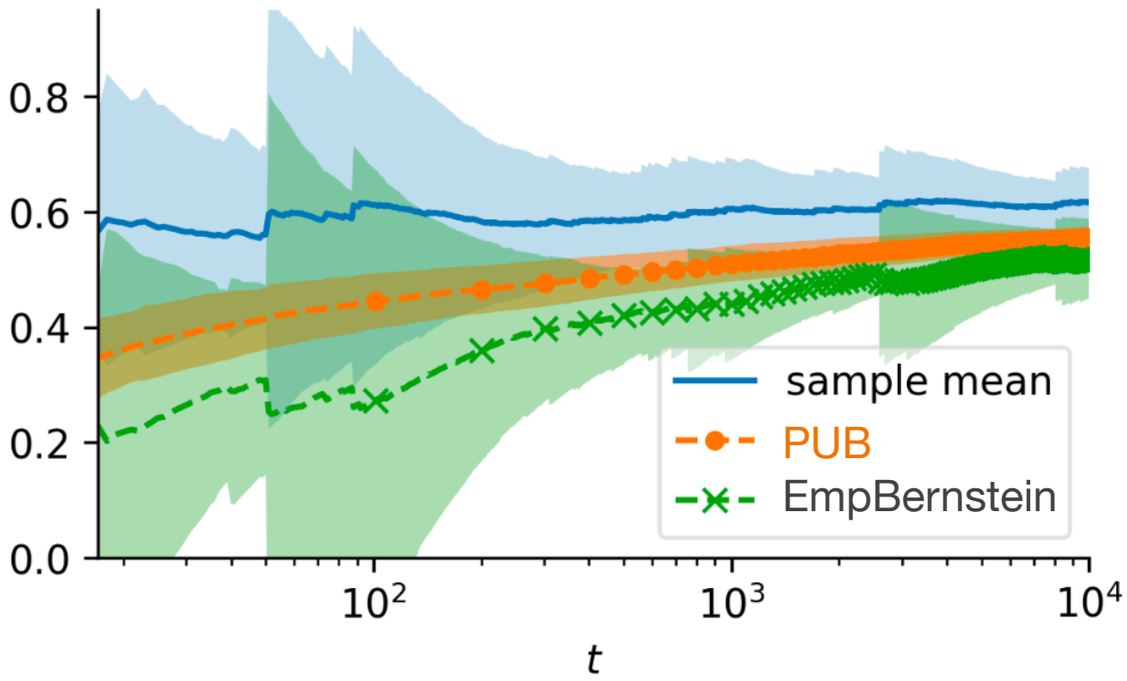
[Kuzborskij**J**WJO'24, Orabona**J**'24, LeeY**J**'24, **Jun**K'24, Jang**J**KO'23, **Jun**BNW'17]

Experiments

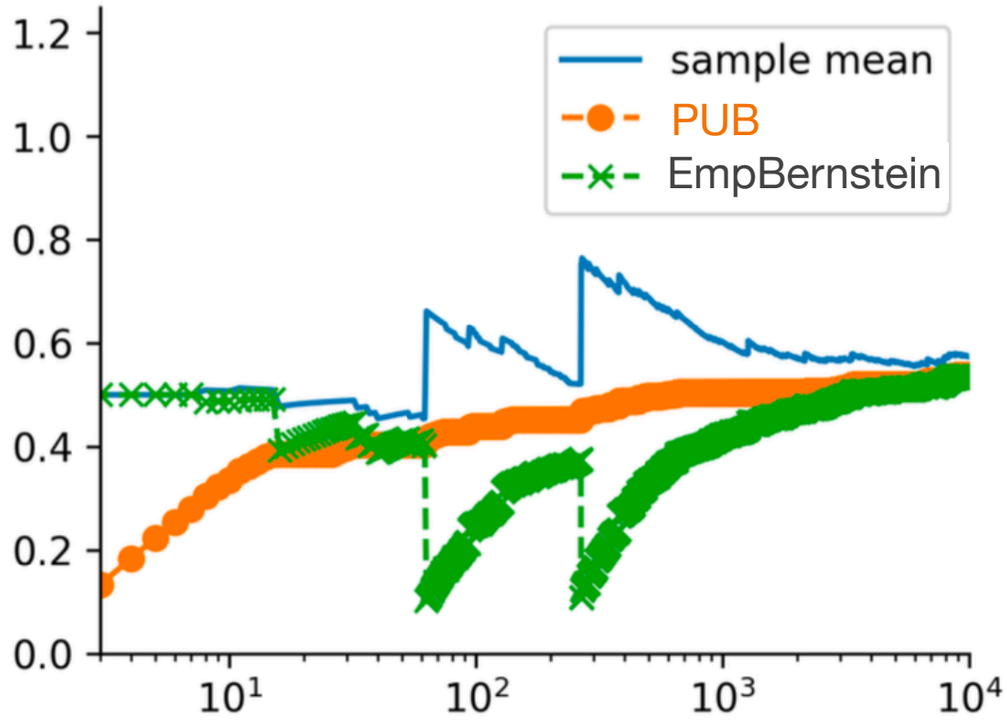
100 runs with X_t with an infinite 4th moment.

EmpBernstein: Plugin variance (heuristic)

average behavior



a specific trial



Selection: Experiments

Relative improvement from ‘Naive’ (higher = better)

Dataset	PenDigits			SatImage			JPVowel		
Size	0.01	0.1	1	0.01	0.1	1	0.01	0.1	1
LS	34.93	**23.60**	3.02	30.88	26.26	**17.84**	40.60	**27.30**	**0.87**
EB	41.34	**22.67**	**3.77**	33.76	26.79	**17.68**	**44.58**	**27.84**	**1.59**
PUB (ours)	**44.02**	**22.74**	**4.01**	**36.14**	**28.16**	**16.35**	**44.51**	**28.59**	**1.40**

[Sakhi+24] → LS

→ EB

Empirical Bernstein: A heuristic based on the central limit theorem.

* see our paper for the results on the ‘learning’ part

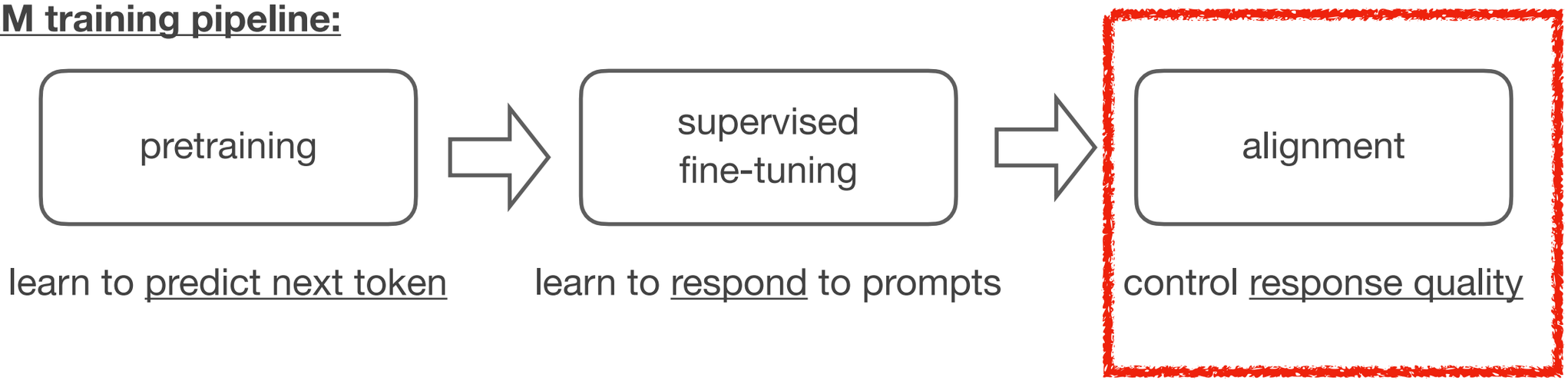
Part 2: LLM Alignment as distribution learning

LLMs can be harmful

- Generate **harmful** content (toxic, biased, unsafe).
  - E.g., how to commit suicide
- Be **unhelpful** (irrelevant, nonsensical, evasive).
- Be **dishonest** or "hallucinate"

Alignment
Make sure the AI agent's behavior align well with human values.

The LLM training pipeline:

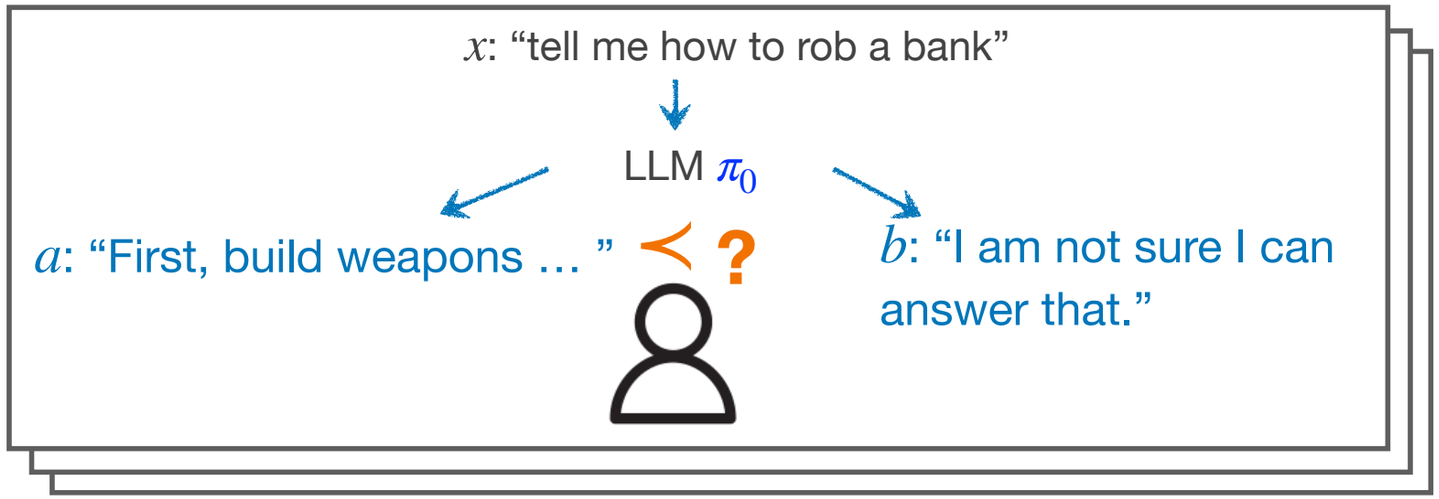


Alignment via pairwise comparison feedback

Input: • A base LLM π_0 (e.g., supervised-fine-tuned)

• A prompt set $\{x\}$

Collect Feedback:



data: $\{(x, a, b, z)\}$

↑

preference: $\mathbf{1}\{a \succ b\}$

or, $\{(x, a_+, a_-)\}$

It's **offline IML**!

	recommendation system	**alignment**
train data	$r(x, a)$ where $a \sim \pi_0(x)$	$a \succ b?$ where $a, b \sim \pi_0(x)$
evaluate on	$a \sim \hat{\pi}(x)$	$a \sim \hat{\pi}(x)$

RLHF (Reinforcement Learning with Human Feedback)

1. Train a reward model $\hat{r}(x, a)$: takes in (x, a) and outputs a score (typically 1-2 layer NN is enough)

Assume: $z \mid x, a, b \sim \text{Bernoulli}\Big(\sigma\big(r(x, a) - r(x, b) \big) \Big)$ // Bradley-Terry model

sigmoid: $\sigma(z) := 1/(1 + e^{-z})$

Use the maximum likelihood estimation on $D_n = \{(x, a, b, z)\}$

2. Optimize reward

$$\max_{\pi \in \Pi} \mathbb{E}_{x \sim D_n, a \sim \pi(x)}[\hat{r}(x, a)] - \beta \cdot \text{KL}(\pi \| \pi_0)$$

↑ do not forget what we learned already in π_0!

hyperparameter

Two main families of alignment algorithms

RLHF
(Reinforcement Learning
from Human Feedback)

1. Learn $\hat{r}(x, a)$ with the MLE

2. $\max_{\pi} \mathbb{E}_{x\sim D_n, a\sim\pi(x)}[\hat{r}(x, a)] - \beta \mathrm{KL}(\pi \| \pi_0)$

makes convergence slow

😱 Slow convergence

😃 Exceptional alignment performance

DPO
(Direct Policy Optimization)

[Rafailov'23]

↑ derived by reformulating RLHF
+ a **strong** assumption

1. $\max_{\pi} \mathbb{E}_{(x,a_+,a_-)\sim D_n} \left[\log \sigma \left(\beta \frac{\pi(a_+|x) \;/\; \pi_0(a_+|x)}{\pi(a_-|x) \;/\; \pi_0(a_-|x)} \right) \right]$

😃 Quick convergence

😱 Worse alignment performance

Note: There are tons of variations of RLHF and DPO

One commonality of prior art

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},a\sim\pi(x)}[\hat{r}(x,a)] - \beta \mathrm{KL}(\pi \| \pi_0)$$

Most existing works take RLHF as starting point.

Even theoretical works take it as the **god-send, untouchable objective** to which they show convergence.

Why is RLHF the right training objective?

the problem definition of alignment does not involve reward at all!

Another problem: **RLHF's asymptotic dilemma**

1. Fix $\beta > 0$: Underfitting

2. Take $\beta = \beta_n \downarrow 0$: Degeneracy ($\hat{\pi}$ is deterministic)

Research question

↓ the

Under what assumptions can we justify e

lear

e.g., pairwise comparison

Beyond RLHF: A Unified Theoretical Framework of Alignment

Jihun Yun, Krafton AI
Juno Kim, Krafton AI intern (⇒ UCB PhD)
Jongho Park, Krafton AI (⇒ UCB PhD)
Junhyuck Kim, Krafton AI
Jongha Jon Ryu, MIT (⇒ PSU Faculty)
Jaewoong Cho, Krafton AI
Kwang-Sung Jun, POSTECH

[submitted]

Proposed: A mathematical formulation of alignment

1. What are we trying to learn?

Language model $\Rightarrow$ **Distribution learning**!

π^*

Let's aim to learn $\hat{\pi}$ that minimizes the distance from π^*

E.g., minimize $\mathrm{KL}(\pi^* \| \hat{\pi})$

Proposed: A mathematical formulation of alignment

2. How is the pairwise comparison data related to the target language model π^*?

Undisputable fact: the model should assign a higher probability to the preferred response

↓ not new

Our choice : Bradley-Terry model + human preference $\propto \pi^*(a \mid x)^\gamma$ ← new // γ: smoothing param.

$$P(a \succ b \mid x) := \frac{\pi^*(a \mid x)^\gamma}{\pi^*(a \mid x)^\gamma + \pi^*(b \mid x)^\gamma} = \sigma\left(\gamma \cdot \ln \frac{\pi^*(a \mid x)}{\pi^*(b \mid x)}\right)$$

what we used to call reward in RLHF: $r(x, a) \propto \gamma \log \pi^*(a \mid x)$

Recap: Proposed alignment framework

Given: π_0: SFT LLM

D_X: a set of prompts $\sim \mathscr{D}$

$D_n = \{(x, a, b, z)\}$

$\in D_X$ $\sim \pi_0(x)$ $\sim$ our_assumption(π^*, x, a, b)

Π: Policy class // e.g., transformer architecture

Output: $\hat{\pi} \in \Pi$

Want: $\mathbb{E}_{x \sim D_X}[\mathrm{KL}(\pi^*(x) \| \hat{\pi}(x))]$ to be small

Assume: realizability $\pi^* \in \Pi$

Method 1: PMLE (Preference MLE)

From our assumption, the MLE is:

$$\hat{\pi} = \arg\min_{\pi \in \Pi} \frac{1}{n} \sum_{(x, a_+, a_-) \in D_n} -\ln\left(\sigma\left(\gamma \ln \frac{\pi(a_+ \mid x)}{\pi(a_- \mid x)}\right)\right) + \beta \cdot \mathrm{KL}(\pi \| \pi_0)$$

(added KL regularizer)

Theorem 1. $\mathbb{E}_{x \sim \mathcal{D}}\left[\mathrm{KL}(\pi^*(x) \,\|\, \hat{\pi}(x))\right] \lesssim C_\Pi \cdot \frac{\ln(|\Pi|/\delta)}{n}.$

Coverage coefficient

$$C_\Pi := \max_{\pi \in \Pi} \frac{\mathbb{E}_{x \sim \mathcal{D},\, a \sim \pi^*(x)}\left[\left(\ln \pi^*(a \mid x) - \ln \pi(a \mid x)\right)^2\right]}{\mathbb{E}_{x \sim \mathcal{D},\, a \sim \pi_0(x)}\left[\left(\ln \pi^*(a \mid x) - \ln \pi(a \mid x)\right)^2\right]}.$$

- At least 1
- Large in general
- Measures the quality of π_0

*some mild assumptions omitted here

PMLE is almost DPO

Ours

$$\arg\min_{\pi\in\Pi} \frac{1}{n}\sum_{i=1}^{n} -\ln\left(\sigma\left(\gamma \ln \frac{\pi(a_i^+ \mid x_i)}{\pi(a_i^- \mid x_i)}\right)\right) + \beta\,\mathsf{KL}(\pi(x_i)||\pi_0(x_i))$$

DPO

$$\arg\min_{\pi\in\Pi} \frac{1}{n}\sum_{i=1}^{n} -\ln(\sigma(\beta \ln(\frac{\pi(a_i^+ \mid x_i)}{\pi_0(a_i^+ \mid x_i)} \cdot \frac{\pi_0(a_i^- \mid x_i)}{\pi(a_i^- \mid x_i)})))$$

Derivation

simple

↑

Assume our preference model → MLE

complicated

↑

RLHF → all-policy assumption → closed-form solution → implicit reward → plug it in to reward model

Method 2: Reverse KL

Note: we want to do distribution learning! ⇒ **Natural objective**: minimize KL divergence

$$\hat{\pi} = \arg\min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}[\mathrm{KL}(\pi(x)\|\pi^*(x))]$$

Challenge: we do not know π^*!

As in RLHF, train a reward model $\hat{R}(x,a)$

⇒ Our assumption says: $\hat{R}(x,a)$ is an estimate of $\gamma \ln \pi^*(a\,|\,x)$

⇒ $\pi^*(a\,|\,x)$ can be estimated by $\pi^*(a\,|\,x) \approx \frac{1}{Z}\exp\left(\frac{1}{\gamma}\hat{R}(x,a)\right)$ ⇒ **plug in!**

$$\arg\max_{\pi} \frac{1}{n}\sum_{(x,\cdot,\cdot)\in D_n}\left[\mathbb{E}_{a\sim\pi(x)}\left[\hat{R}(x,a)\right] + \gamma H(\pi(x))\right] - \boxed{\beta\cdot\mathrm{KL}(\pi\|\pi_0)}$$

added regularizer

setting $\gamma = 0$ recovers the original RLHF objective!

⇒ strict generalization of RLHF

Method 2: Reverse KL (≈ RLHF)

Theorem 2. $\mathbb{E}_{x\sim\mathcal{D}}\left[\mathrm{KL}(\pi^*(x)\,\|\,\hat{\pi}(x))\right] \lesssim \frac{\ln(|\Pi|/\delta)}{n} + C_{\mathscr{R}}\cdot\frac{\ln(|\mathscr{R}|/\delta)}{n}.$

smaller! also smaller!

For the first time, we provide a theoretical justification for RLHF, with a minor correction!

Compare with PMLE (≈ DPO):

Theorem 1. $\mathbb{E}_{x\sim\mathcal{D}}\left[\mathrm{KL}(\pi^*(x)\,\|\,\hat{\pi}(x))\right] \lesssim C_{\Pi}\cdot\frac{\ln(|\Pi|/\delta)}{n}.$

For the first time, we theoretically confirmed the empirical finding that RLHF is better than DPO!

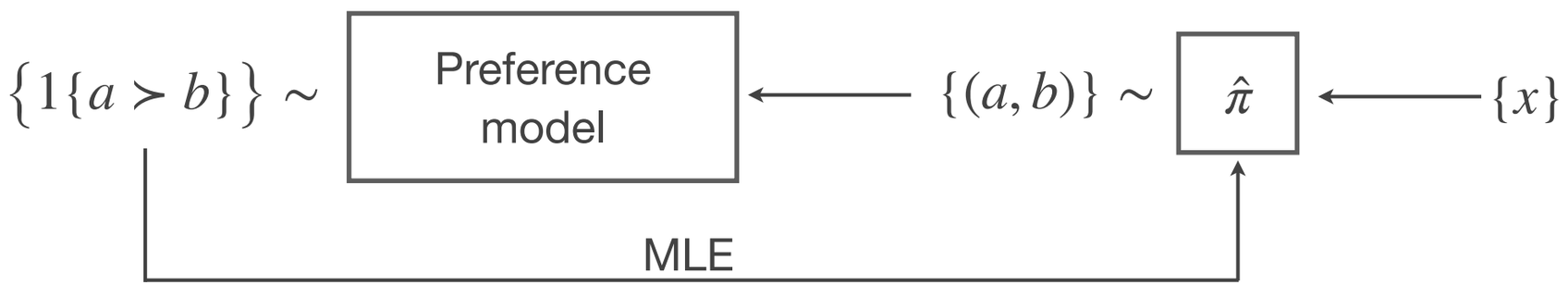
Method 3: Find a middleground between RLHF and DPO

Generic observation: Training a reward model + onpolicy sampling works well.

	RLHF	DPO	???
uses reward model	Y	N	Y
ease of optimization (likelihood-based objective)	N	Y	Y

Proposed: Preference distillation

Key: "reward model = preference model" ⇒ we can simulate the human feedback!



Experiments: Summarization task

Model	Algorithm	Win-rate(↑)	RM score(↑)	$\mathrm{KL}(\pi \| \pi_0)(\downarrow)$
2.8B	DPO	47.3±1.01	**2.66**±0.03	64.35±0.68
	PMLE	**49.1**±1.43	2.38±0.04	**31.42**±0.79
	REBEL	71.0±1.26	2.86±0.02	**25.67**±0.61
	Pref. Distill.	**73.8**±1.47	**2.91**±0.03	28.38±0.82
	RLHF	72.0±0.67	2.95±0.05	**24.94**±1.12
	Reverse KL	**73.0**±0.93	**3.03**±0.04	25.91±0.94
6.9B	DPO	55.6±0.53	6.09±0.02	54.45±0.82
	PMLE	**59.0**±0.74	**6.27**±0.03	**24.20**±0.92
	REBEL	82.5±0.93	7.64±0.02	29.80±0.71
	Pref. Distill.	**83.9**±0.59	**7.66**±0.02	**29.05**±1.02
	RLHF	82.0±0.92	**7.85**±0.03	24.38±0.77
	Reverse KL	**84.0**±1.01	7.77±0.01	**24.12**±0.63

winrate: vs human reference summary, judged by GPT-4

Future work: IML problems in LLMs and beyond

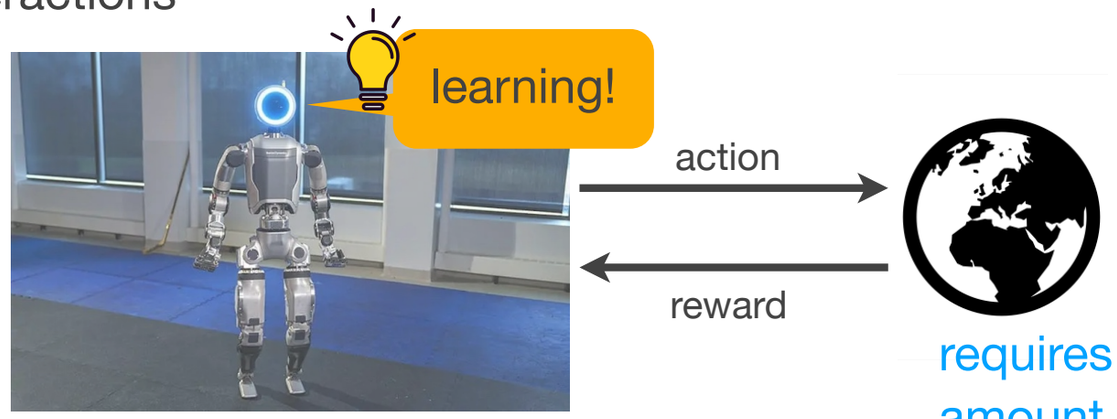


- How to explore efficiently in test time?

- Prevent over-optimization in IML problems

Offline IML is more important than ever

Traditionally… assumes real-world interactions



requires an unrealistic amount of time & cost!

Nowadays … use ML-based simulators to obtain synthetic data for training

e.g., Distillation — LLMs to generate synthetic data
Reward models for LLM post-training
In-silico drug discovery

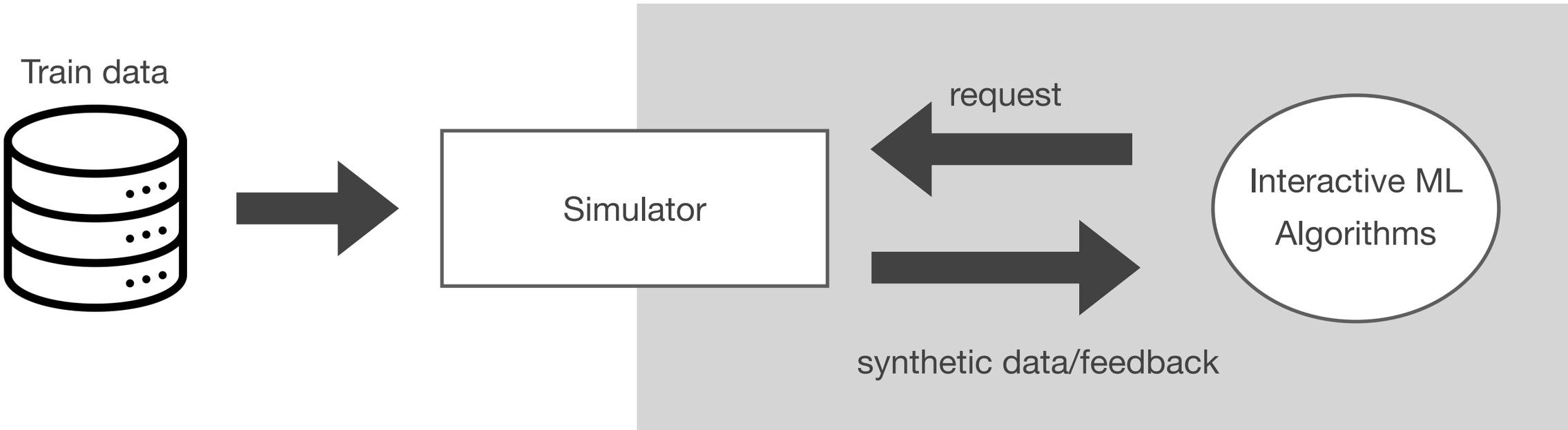
These are **offline IML**
⇒ Potentially shares the same issue: Over-optimization!

Max Welling



"Synthetic data is 5th industrial revolution" @NeurIPS'23

Learned simulators



E.g., In-silico drug discovery

Materials discovery

ML-based PDE solvers

Imperfect simulator causes over-optimization..

Can we establish a universal framework to mitigate over-optimization with simulators generating feedback beyond scalar rewards?

Summary



Offline interactive ML is becoming everywhere
but we just started understanding it!